

# Yuan Chang (Merlin)

Portfolio: [itis2010me.github](https://github.com/itis2010me)

Github: [github.com/itis2010me](https://github.com/itis2010me)

Email: [merlin.y.chang@gmail.com](mailto:merlin.y.chang@gmail.com)

Mobile: +1 (530) 760 6690

## EDUCATION

- **University of California, San Diego** La Jolla, California  
*Master of Science - Computer Science and Engineering* 2022 - 2024
- **University of California, Davis** Davis, California  
*Bachelor of Science - Computer Science; Graduated with Honors* 2018 - 2022  
*Minor: Mathematics*

## EXPERIENCES & INTERNSHIPS

- **Amazon Web Services (AWS)** Seattle, WA  
*Software Developer Engineer* Apr 2024 - **Present**
- **Amazon AWS** Seattle, WA  
*Software Developer Engineer Intern (High Performance Computing group)* June 2023 - Sep 2023
  - Designed and implemented internal Restful APIs that configure computing instances with cluster management and job scheduling systems.
  - Developed public API within AWS CLI to enable direct interactions of our service for customers.
  - Developed database managers which queried a single-table DynamoDB storing both cluster and node level data.
  - Simplified dependency graph within the team project and incorporated dependency injections to handle complex and nested object-oriented codebase.
  - Implemented API orchestration that simplified complex distributed workloads into a simple and seamless API invocation.
  - Incorporated unit, integration and load tests into the CI/CD production pipeline which achieved over 85% test coverage to ensure workflow validity.
- **UC Davis Applied Mathematics Research Group** Davis, CA  
*Research Assistant & Software Engineer* June 2021 - June 2022
  - Developed automated reasoning pipeline that computed undiscovered Ramsey constants.
  - Utilized symmetric breaking and optimization techniques to reduce the search space by 6x with preprocessing.
  - Optimized the computation to achieve up to 10x improvements in source generation.
  - Leveraged parallel computing to speedup SAT computation time by 2x.
  - Research paper and artifacts published on *ACM (ISSAC 2022)*.

## PROJECTS

- **Minimization of CA Wildfire Impact(Web Mining, Convex Optimization, KKT, Lagrangian, Data Analysis):** Constructed connectivity matrix of CA counties based on humidity and fire probability. Formulated the CA wildfire problem into a constraint optimization problem and solved through integer programming techniques. Computed counties with highest priorities based on 2021's historical data, while minimizing each counties' financial, social and geological impact.(2023)
- **Terminal UI Board-game In Haskell(Haskell, Functional Programming, QuickCheck, Genetic Algorithm):** Implementing the TUI version of the classic board-game, Mastermind, in Haskell. Monadic and functional programming. Implemented 10 QuickCheck property testings. Implemented genetic algorithm-based AI player using minimax techniques. Github Link (2022)
- **Computations with Rado Numbers and Degree of Regularity(Automated Reasoning, Theory of Computation, Number Theory, Combinatorics):** Advancement in terms of Rado Numbers and the degree of regularity. Discovery of nearly 500 new mathematical constants. Results collected and formulated into research paper accepted into 2022 ISAAC conference. Co-author with Prof. Jesús De Loera and William Wesley. Github Link (2021)

## PUBLICATIONS

- **Rado Numbers and SAT Computations (with J. A. De Loera and W. J. Wesley).** Proceedings of the 47th International Symposium on Symbolic and Algebraic Computation (ISSAC 2022). Pages 333-342, available online at <https://dl.acm.org/doi/10.1145/3476446.3535494>.

## ACADEMIA EXPERIENCE

- **Graduate Teaching Assistant (Halicioğlu Data Science Institute UCSD)** La Jolla, California  
*DSC 211 - Introduction to Optimization.* Jan 2024 - Mar 2024

## TECHNICAL SKILLS SUMMARY

- **Programming Languages:** Java, Python, Typescript, C++, Bash, LEAN
- **Libraries/Tools:** Git, pdb, UNIX, SAT/SMT solvers, TensorFlow, pandas, NumPy, SymPy
- **Cloud service/Model/Interface:** AWS, SaaS, IaaS, Restful API, DynamoDB